UNITED ARAB EMIRATES MINISTRY OF EDUCATION
الإمارات العربية المتحدة وزارة التربية والتعليم

Al Khalidiya Boys School – Cycle 3

School code 10048 | Dibba Al Hisn | Sharjah Eastern

Grade 12 Physics Final Exam Practice

Term 1 2026–2027 • Set A • With Answers

100 marks | 150 minutes | Calculator allowed

Answer all 20 questions. Choose one option for each MCQ. Show working, units and labelled diagrams for Q17–20. Use g = 9.8 m/s² when needed.

Section A Multiple Choice 60 marks

1. A 20 kg bag is lowered at constant speed through 1.2 m. What work does gravity do? Use g = 9.8 m/s². [4 marks]

A. +24 J
B. −235 J
C. +235 J
D. 0 J

Reference: p. 20, Practice Problem 31 | **Answer C**

2. The elastic potential energy stored in a spring is: [4 marks]

A. ½kx²
B. mgh
C. ½mv²
D. kx

Reference: pp. 20–22 | **Answer A**

3. In a perfectly inelastic collision, the colliding objects: [4 marks]

A. Always lose all momentum
B. Stick together and share a final velocity
C. Always conserve kinetic energy
D. Cannot move after impact

Reference: pp. 29–31 | **Answer B**

4. As an object's temperature rises, the peak wavelength of its thermal radiation generally: [4 marks]

A. Becomes infinite
B. Increases
C. Stays fixed for all temperatures
D. Decreases

Reference: pp. 46–48 | **Answer D**

5. A photon has wavelength 515 nm. Using hc = 1240 eV nm, its energy is: [4 marks]

A. 515 eV
B. 6.39 × 10⁵ eV
C. 2.41 eV
D. 0.415 eV

Reference: p. 52, Practice Problem 1 | **Answer C**

6. The threshold frequency is the minimum light frequency needed to: [4 marks]

A. Eject electrons from a specified metal
B. Make any light visible
C. Produce interference
D. Change every atom into an ion

Reference: pp. 49–51; p. 58 | **Answer A**

7. At a fixed frequency above threshold, increasing light intensity mainly increases the: [4 marks]

A. Work function
B. Photon energy
C. Maximum kinetic energy of each electron
D. Number of emitted electrons per second

Reference: pp. 51–56 | **Answer D**

8. A stopping potential of 4.0 V corresponds to a maximum photoelectron kinetic energy of: [2 marks]

A. 0.25 eV
B. 0 eV
C. 4.0 eV
D. 4.0 J

Reference: p. 54, Example Problem 1 | **Answer C**

9. Photon momentum is given by: [4 marks]

A. $p = \lambda/h$
B. $p = h\lambda$
C. $p = \frac{1}{2}m_{ec}^2$
D. $p = h/\lambda$

Reference: pp. 57–58 | **Answer D**

10. The de Broglie wavelength of a moving particle is: [4 marks]

A. $\lambda = mc^2$
B. $\lambda = p/h$
C. $\lambda = h/p$
D. $\lambda = hp$

Reference: pp. 59–60 | **Answer C**

11. Heisenberg's uncertainty principle limits simultaneous precise knowledge of a particle's: [4 marks]

A. Position and momentum
B. Charge and colour
C. Energy and mass only
D. Mass and charge

Reference: p. 61, Check Your Progress 32 | **Answer A**

12. Rutherford's rare large-angle alpha-particle deflections showed that an atom has: [4 marks]

A. No empty space
B. A small, dense, positively charged nucleus
C. Positive charge spread uniformly through all its volume
D. A nucleus larger than the whole atom

Reference: pp. 62–63; p. 74, Check Your Progress 41 | **Answer B**

13. Why can an emission-line spectrum identify a gas? [4 marks]

A. The spectrum depends only on gas volume
B. It contains every wavelength equally
C. Every gas has identical energy levels
D. Its allowed energy differences produce a characteristic set of lines

Reference: pp. 64–66; p. 74, Check Your Progress 40, 42–44 | **Answer D**

14. In the Bohr model, an electron in an allowed stationary orbit: [4 marks]

A. Can have any arbitrary orbital energy
B. Does not continuously radiate energy
C. Continuously spirals into the nucleus
D. Must emit a photon every revolution

Reference: pp. 67–69 | **Answer B**

15. For hydrogen $E_n = -13.6/n^2$ eV. What is E_2? [4 marks]

A. −6.80 eV
B. −3.40 eV
C. +3.40 eV
D. −27.2 eV

Reference: pp. 69–71; p. 74, Check Your Progress 46 | **Answer B**

16. Hydrogen transitions ending at n = 1 form the: [2 marks]

A. Lyman series
B. Paschen series
C. Balmer series
D. Continuous spectrum

Reference: pp. 72–73, 75–79 | **Answer A**

Set A Section B Free Response

17. A bicycle and rider have combined mass 85.0 kg and approach a hill at 8.5 m/s. They coast uphill without friction until momentarily at rest. Take gravitational potential energy as zero at the starting height and ignore wheel rotation for this model. [10 marks]

Reference: p. 28, Practice Problems 39–40; pp. 23–25

(a) Calculate the initial translational kinetic energy. [2]

Answer $KE = \frac{1}{2}mv^2$ [1] $= \frac{1}{2}(85.0)(8.5)^2 = 3071$ J [1].

(b) State the energy equation and calculate the maximum height gained. [3]

Answer $\frac{1}{2}mv^2 = mgh$ [1]; $h = v^2/(2g)$ [1] = 3.69 m [1].

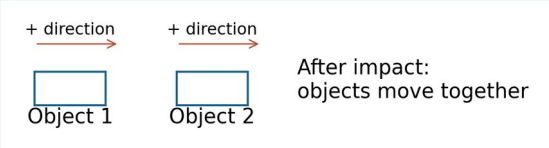
(c) Find the work done by gravity while climbing to that height. [2]

Answer $W_g = -\Delta PE$ [1] = −3071 J [1].

(d) Explain how the result would change if friction acted and where the lost mechanical energy would go. [3]

Answer Maximum height would be lower [1]; some initial KE becomes thermal/internal energy [1]; total energy including surroundings remains conserved [1].

18. A 1150 kg car travelling at +15.0 m/s collides with a 1575 kg car travelling at +5.0 m/s. They lock together. Neglect external impulse. [10 marks]



Reference: p. 31, Example Problem 6

(a) Find the initial total momentum. [2]

Answer $p_i = 1150(15) + 1575(5)$ [1] = 25125 kg m/s [1].

(b) Calculate the common velocity after collision. [2]

Answer $v_f = p_i/(1150 + 1575)$ [1] = +9.22 m/s [1].

(c) Calculate initial and final kinetic energy, and the decrease. [4]

Answer $KE_i = ½(1150)(15^2) + ½(1575)(5^2)$ = 149062.5 J [1]; $KE_f = ½(2725)(9.22018^2)$ [1] = 115828.6 J [1]; decrease = 33233.9 J [1].

(d) Classify the collision and explain what happened to the missing kinetic energy. [2]

Answer Perfectly inelastic [1]; converted to deformation, thermal energy and sound [1].

19. Zinc has threshold wavelength 310 nm. Use $c = 3.00 \times 10^8$ m/s, hc = 1240 eV nm and $h = 4.14 \times 10^{-15}$ eV s. [10 marks]

Reference: p. 56, Practice Problem 12; p. 55, Figure 9 and Connecting Math to Physics

(a) Find the threshold frequency. [2]

Answer $f_0 = c/\lambda_0$ [1] = 9.68×10^{14} Hz [1].

(b) Find the work function in eV. [2]

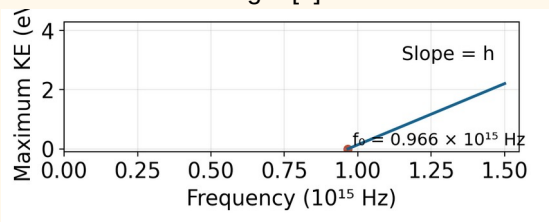
Answer $\varphi = hc/\lambda_0$ [1] = 4.00 eV [1].

(c) Sketch KE_{max} versus frequency, labelling the threshold and slope. Write its equation with these constants. [3]

Answer Straight line starting at (9.68×10^{14} Hz,0), labelled axes [1]; slope h [1]; $KE_{max}(eV) = (4.14 \times 10^{-15})f - 4.00$ [1].

(d) Can 425 nm light eject electrons from zinc? Explain, and state how increasing intensity would affect your conclusion. [3]

Answer No [1]; photon energy 1240/425 = 2.92 eV < 4.00 eV [1]; greater intensity still cannot overcome the threshold at this wavelength [1].



20. In hydrogen, $E_n = -13.6/n^2$ eV and $r_n = n^2(0.053$ nm). An electron is excited from n = 1 to n = 2 and subsequently returns to n = 1. Use $h = 4.14 \times 10^{-15}$ eV s and hc = 1240 eV nm. [10 marks]

Reference: p. 72, Example Problem 4 and Practice Problem 36; p. 73, Example Problem 5

(a) Calculate E_1, E_2 and the energy absorbed in excitation. [3]

Answer $E_1 = -13.6$ eV [1]; $E_2 = -3.40$ eV [1]; $\Delta E = +10.2$ eV absorbed [1].

(b) Find frequency and wavelength of the photon emitted on return. [3]

Answer $E_{photon} = 10.2$ eV [1]; $f = E/h = 2.46 \times 10^{15}$ Hz [1]; $\lambda = hc/E = 122$ nm [1].

(c) Calculate r_2 and identify the spectral series and region of this emission. [2]

Answer $r_2 = 4(0.053) = 0.212$ nm [1]; Lyman series, ultraviolet [1].

(d) Draw a labelled two-level energy diagram with arrows for absorption and emission. [2]

Answer Correct E_1 and E_2 levels [1]; upward absorption and downward emission arrows [1].

